

INDUSTRIAL INTERNSHIP REPORT

"Prediction of Agriculture Crop Production in India"

Prepared by

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Internship Domain	Data Science & Machine Learning
Organisation	UpSkill Campus & UniConverge Technologies Pvt. Ltd.
Internship Period	17 June 2026 – July 2026 (6 Weeks)
Mode	Remote
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Executive Summary

This report documents a 6-week Industrial Internship in Data Science & Machine Learning under UpSkill Campus and U

TABLE OF CONTENTS

1. Preface	3
2. Introduction	4
2.1 About UniConverge Technologies Pvt. Ltd.	4
2.2 About UpSkill Campus (USC)	4
2.3 Objectives	4
2.4 References	4
2.5 Glossary	5
3. Problem Statement	6
4. Existing and Proposed Solution	7
4.1 Existing Solutions and Their Limitations	7
4.2 Proposed Solution	7
4.3 Code Submission (GitHub Link)	7
4.4 Report Submission (GitHub Link)	7
5. Proposed Design / Model	8
5.1 High Level Pipeline Diagram	8
5.2 Low Level Design — Model Architecture	8
5.3 Interfaces — Streamlit Web Application	8
6. Performance Test	10
6.1 Test Plan / Test Cases	10
6.2 Test Procedure	10
6.3 Performance Outcome	10
7. My Learnings	11
8. Future Work Scope	12

1. Preface

This report presents the complete work carried out during a six-week Industrial Internship programme offered by UpSkill Campus in collaboration with UniConverge Technologies Pvt. Ltd. The internship was conducted remotely in the domain of Data Science and Machine Learning, commencing 17 June 2026.

The project undertaken was titled '**Prediction of Agriculture Crop Production in India**' — a regression-based machine learning project using real historical government data covering 29 Indian states, 124 crop types, and 6 seasons across the years 1997 to 2015 (246,091 records).

Over six weeks, the complete data science pipeline was implemented independently: dataset acquisition, data cleaning and quality validation, exploratory data analysis (EDA), feature engineering, training and comparing three machine learning models, building a deployed Streamlit web application, and maintaining a clean GitHub repository. Each week's progress was submitted as a formal progress report on the UpSkill Campus platform.

This internship provided hands-on exposure to real-world data quality issues, model selection methodology, deployment constraints, and professional engineering practices that go well beyond typical academic coursework. I am grateful to the UpSkill Campus team and UniConverge Technologies for creating this structured learning opportunity.

To fellow students and juniors: theoretical knowledge and practical application are two very different skills. Start building real projects early — you will learn more from one messy real-world dataset than from ten textbook examples. This internship proved that.

2. Introduction

2.1 About UniConverge Technologies Pvt. Ltd.

UniConverge Technologies Pvt. Ltd. (UCT), established in 2013, is a digital transformation company providing industrial solutions with a focus on sustainability and return on investment. UCT leverages cutting-edge technologies including IoT, Cyber Security, Cloud Computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, and front-end frameworks.

- **UCT Insight (IoT Platform)** — Designed for quick deployment of IoT applications, built in Java and ReactJS, supporting MQTT, CoAP, HTTP, Modbus TCP, and OPC UA. Features include custom dashboards, analytics, alerting, and third-party integration (Power BI, SAP, ERP).
- **Smart Factory Platform** — Scalable solution for production and asset monitoring, OEE measurement, predictive maintenance, and digital twin capabilities.
- **LoRaWAN Solutions** — Solutions in AgriTech, Smart Cities, Industrial Monitoring, Smart Street Lighting, and Smart Metering.
- **Predictive Maintenance** — Industrial machine health monitoring leveraging Embedded Systems, Industrial IoT, and Machine Learning.

2.2 About UpSkill Campus (USC)

UpSkill Campus (USC), in association with UniConverge Technologies, facilitated the smooth execution of this internship programme. USC is a career development platform that delivers personalised executive coaching in an affordable, scalable, and measurable way. USC aims to upskill one million learners over the next five years and provides services including structured internships, industry expert interactions, project-based learning, and career growth support.

2.3 Objectives

- Gain practical, hands-on experience working on an industry-relevant data science problem.
- Solve a real-world prediction problem using machine learning techniques on real data.
- Train and compare multiple models using a scientific, evidence-based approach to model selection.
- Deploy the best-performing model as a user-facing interactive web application.
- Develop professional engineering habits: version control (Git/GitHub), structured documentation, and modular code organisation.
- Build a thorough understanding of the complete ML workflow — from raw data to a deployed solution.

2.4 References

- Kaggle — Crop Production in India Dataset:
<https://www.kaggle.com/datasets/abhinand05/crop-production-in-india>

- scikit-learn Documentation: <https://scikit-learn.org>
- Streamlit Documentation: <https://docs.streamlit.io>
- pandas Documentation: <https://pandas.pydata.org>
- joblib Documentation: <https://joblib.readthedocs.io>
- GitHub Repository: <https://github.com/jiralikarjagruti-svg/upskillCampus>

2.5 Glossary

Term / Acronym	Definition
ML	Machine Learning
EDA	Exploratory Data Analysis — visualising and understanding raw data
R ² Score	Coefficient of Determination — model accuracy metric (0 to 1, higher = better)
MAE	Mean Absolute Error — average prediction error in tonnes (lower = better)
Random Forest	Ensemble ML model using multiple decision trees to improve accuracy
Label Encoding	Converting categorical text values into numeric format for ML models
Streamlit	Python library for building interactive data science web applications
joblib	Python library for saving/loading ML models efficiently
GitHub	Cloud-based platform for hosting and version-controlling code
Kharif	Indian monsoon cropping season (June – October)
Rabi	Indian winter cropping season (October – March)

3. Problem Statement

Agriculture is the backbone of India's economy, contributing approximately 18% to GDP and employing over 50% of the workforce. Accurately forecasting crop production is critical for food security planning, supply-chain management, price stabilisation, and agricultural policy-making.

However, crop production is influenced by a complex combination of factors — geographic region, crop variety, cultivated area, seasonal weather patterns, and year-on-year variations — making accurate prediction a non-trivial problem.

The assigned problem was to:

- Build a machine learning regression model capable of predicting the total production (in tonnes) of a specified crop in a given Indian state.
- Train the model on real government historical data covering 29 states, 124 crop types, and 6 seasons from 1997 to 2015 (246,091 records).
- Accept the following inputs: State Name, Crop Year, Season, Crop Type, and Cultivated Area (hectares).
- Output a predicted Production value (in tonnes) with an estimated confidence range.
- Deliver the solution as an interactive web application accessible to non-technical users.

An additional real-world data quality challenge was encountered: the dataset contained unit inconsistencies (Coconut production in nuts/count vs. tonnes for all other crops) and an incomplete year (2015: only 561 records vs. ~13,000 average). Identifying and resolving these issues was part of the project scope.

4. Existing and Proposed Solution

4.1 Existing Solutions and Their Limitations

- **Simple Linear Regression:** Assumes a linear relationship between area and production. In practice, this is highly non-linear. Confirmed in this project: Linear Regression achieved only $R^2 = 0.074$ (7.4%) on this dataset.
- **Statistical time-series models:** Focus only on year-over-year trends and ignore the categorical variables (crop type, season, state) that are the strongest predictors.
- **No user-facing interface:** Most solutions are notebooks/scripts inaccessible to non-technical agricultural officers or policymakers.
- **Data quality blind spots:** Published analyses rarely address unit mismatches or incomplete yearly records, which distort analysis results if unaddressed.

4.2 Proposed Solution

The proposed solution is a complete end-to-end ML pipeline using a **Random Forest Regressor** — an ensemble model that captures complex non-linear relationships and handles mixed data types (categorical + numerical) effectively.

- Rigorous data cleaning to handle missing values, whitespace inconsistencies, unit mismatches, and incomplete yearly records.
- Label Encoding to convert categorical features (State, Season, Crop) into numeric inputs.
- Systematic training and comparison of three regression models on identical splits, evaluated by R^2 score and MAE.
- A professional Streamlit web application with three tabs (Predict / Historical Trends / Model Comparison) usable by non-technical stakeholders.
- All code, data, and model files hosted on a public GitHub repository with version control.

4.3 Code Submission (GitHub Link)

<https://github.com/jiralikarjagruti-svg/upskillCampus>

4.4 Report Submission (GitHub Link)

https://github.com/jiralikarjagruti-svg/upskillCampus/blob/main/Jagruti_Jiralikar_InternshipReport.pdf

5. Proposed Design / Model

5.1 High Level Pipeline Diagram

Stage	Activity	Output / Deliverable
1. Data Acquisition	Downloaded dataset from Kaggle	crop_production.csv (246,091 rows, 7 cols)
2. Data Cleaning	Removed missing rows; Fixed whitespace in Crop names	242,361 clean records
3. EDA	Analysed crops, states, seasons, year trends, correlations	5 charts; 2 data quality findings
4. Feature Engineering	Label Encoding; 80/20 train-test split	X_train (192,322 rows) X_test (48,081 rows)
5. Model Training	Trained & compared 3 models	Best: Random Forest R ² =0.958, MAE=8,638 t
6. Model Saving	joblib compress=3	crop_model.pkl (17 MB) Reduced from 1.5 GB
7. Web Application	Streamlit: 3 tabs, custom dark theme	Interactive prediction tool
8. GitHub Push	Version controlled all files with commit messages	Public repository live

5.2 Low Level Design — Model Architecture

Component	Details
Algorithm	Random Forest Regressor (ensemble of 50 decision trees)
n_estimators	50 (reduced from 100 to optimise saved file size)
max_depth	15 levels per tree (controls model complexity / overfitting)
random_state	42 (ensures full reproducibility across all runs)
Input Features (5)	State_encoded, Crop_Year, Season_encoded, Crop_encoded, Area
Target Variable	Production (in tonnes)
Training Set	192,322 rows (80% of 240,403 usable records)
Test Set	48,081 rows (20% of usable records)
Serialisation	joblib.dump(model, 'crop_model.pkl', compress=3) → 17 MB

5.3 Interfaces — Streamlit Web Application

- **Tab 1 — Predict:** Dropdown selectors for State, Season, Crop; numeric inputs for Year and Area. Returns predicted production, confidence range (from 50 individual tree predictions), input summary table, and CSV download button. Feature importance chart permanently visible below.
- **Tab 2 — Historical Trends:** Interactive line chart of year-wise production for any selected state-crop combination (1997–2015), with average, maximum, and record count metrics.
- **Tab 3 — Model Comparison:** Side-by-side bar charts comparing R^2 and MAE across all three models, a detailed results table, and an explanatory insight note.

Custom dark theme with agriculture green accents (#4CAF50) via `.streamlit/config.toml`. Sidebar includes developer credits (Jagruti Jiralikar, Data Science & ML Intern @ UpSkill Campus) and a copyright footer.

6. Performance Test

6.1 Test Plan / Test Cases

- **Same dataset split:** 80% training (192,322 rows) and 20% testing (48,081 rows), random_state=42 for reproducibility.
- **Same input features:** State_encoded, Crop_Year, Season_encoded, Crop_encoded, Area.
- **Same metrics:** R² Score (primary) and Mean Absolute Error — MAE (secondary).
- **Three models tested:** Linear Regression, Decision Tree Regressor, Random Forest Regressor.

6.2 Test Procedure

- Each model fitted on X_train, y_train with specified parameters.
- Predictions generated on X_test using model.predict(X_test).
- R² Score calculated using sklearn.metrics.r2_score(y_test, y_pred).
- MAE calculated using sklearn.metrics.mean_absolute_error(y_test, y_pred).
- Feature importance extracted from Random Forest using model.feature_importances_.
- Results stored in a dictionary and exported as model_comparison.json for the web app.

6.3 Performance Outcome

Model	R ² Score	MAE (tonnes)	Assessment
Linear Regression	0.074 (7.4%)	64,592	Poor — high bias on non-linear data
Decision Tree	0.799 (79.9%)	10,717	Good — captures non-linearity
✓ Random Forest	0.958 (95.8%)	8,638	Excellent — selected as final model

Key Findings:

- Random Forest achieved **R² = 0.958** — correctly explaining 95.8% of variance in crop production, well above the industry benchmark of 0.85+.
- Feature importance: **Crop type (60.8%)** is the dominant predictor, followed by Area (26.0%), State (9.3%), Year (2.9%), and Season (1.0%). Crop choice matters more than location or timing.
- Linear Regression's failure (R²=0.074) validates the dataset's non-linear nature, justifying the ensemble approach.
- Model optimisation reduced file size from **1.5 GB to 17 MB** (98.9% reduction) while maintaining identical R² of 0.958 — zero loss in accuracy.
- Two critical data quality issues identified and resolved: (a) Coconut unit mismatch; (b) Incomplete 2015 records (561 rows vs ~13,000 average) — excluded from trend analysis.

7. My Learnings

- **Real-world data is always messy.** The dataset had missing target values, extra whitespace in category names, unit mismatches, and an incomplete final year. Identifying and handling these is as critical as the model — and rarely taught in academic settings.
- **Model selection must be evidence-based.** Comparing three models on identical splits revealed that Linear Regression's simplicity is a liability on non-linear data ($R^2=0.074$), while Random Forest captured complex patterns ($R^2=0.958$). The right model is discovered through experimentation, not assumed.
- **Feature importance turns predictions into insights.** Crop type (60.8%) and Area (26%) account for 87% of prediction power — an actionable agricultural insight, not just a modelling metric.
- **Engineering constraints are real.** A 1.5 GB model file was a deployment blocker. Resolving it through parameter reduction and joblib compression (17 MB, same accuracy) was a practical engineering problem no course covers.
- **Deployment bridges ML and real-world impact.** A model in a notebook is a proof of concept. A model in a professional Streamlit app with tabs, confidence ranges, and download buttons is a tool a real stakeholder could use.
- **Version control is professional hygiene.** Using Git throughout — meaningful commits, .gitignore, clean folder structure — directly mirrors industry practice.
- **Theory and practice reinforce each other.** Studying ensemble methods and the bias-variance tradeoff made the project results more interpretable. Hands-on results made theory intuitive. This bidirectional learning is the core value of project-based internships.

8. Future Work Scope

- **Hyperparameter Tuning:** GridSearchCV or RandomizedSearchCV to optimise `n_estimators`, `max_depth`, `min_samples_split` — potentially pushing R^2 above 0.96.
- **Weather and Soil Data Integration:** Merging with historical rainfall, temperature, and soil quality data would add the strongest real-world predictors of crop yield.
- **District-Level Predictions:** `District_Name` is available in the dataset but unused. Adding district granularity would enable significantly more localised predictions.
- **Crop Recommendation System:** Extending from prediction ('how much will this crop yield?') to recommendation ('which crop should I grow for maximum yield?') — a direct agricultural decision-support tool.
- **Online Deployment:** Deploying to Streamlit Community Cloud or AWS for public access via a permanent URL, without local installation.
- **Time-Series Forecasting:** Replacing regression with LSTM or Prophet to forecast future year production trends — enabling forward-looking agricultural planning.
- **Cross-Validation:** Implementing k-fold cross-validation ($k=5$ or 10) for a statistically more robust estimate of model generalisation performance.

GitHub Repository (Code + Report):
<https://github.com/jiralikarjagruti-svg/upskillCampus>

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